Practice workbook · Binary search step by step

Stepping through binary search on a sorted list (order of items examined, low/high/mid values), worst-case number of checks, and why it fails on unsorted data.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-10

1. Question 5 · part b · 4 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part b only.

5 Programmers refine algorithms, use searches and read data from files.

(a) **Figure 6** shows an algorithm that determines whether the animal name inputted by the user is found.

```
1  SET a1 TO 'alpaca'
2  SET a2 TO 'bear'
3  SET a3 TO 'camel'
4  SET a4 TO 'deer'
5
6  SEND ('Enter an animal: ') TO DISPLAY
7  RECEIVE target FROM (STRING) KEYBOARD
8
9  IF (a1 = target) THEN
10     SEND ('Found') TO DISPLAY
11 ELSE
12     IF (a2 = target) THEN
13         SEND ('Found') TO DISPLAY
14     ELSE
15         IF (a3 = target) THEN
16             SEND ('Found') TO DISPLAY
17         ELSE
18             IF (a4 = target) THEN
19                 SEND ('Found') TO DISPLAY
20             END IF
21         END IF
22     END IF
23 END IF
```

Figure 6

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1. Question 5 · part b · 4 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part b only.

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Explain **one** improvement to this algorithm that will reduce the number of variables required and will enable it to work with any number of animals.

(2)

(b) Binary search is a divide and conquer algorithm.

Here is a list of data.

10	12	13	14	15	16	17	18	19	20

A binary search is used to check whether the number 11 is in the list.

The midpoint is calculated by adding the high index to the low index and **integer dividing** by 2

Complete the table to show the values for the high index, the low index and the midpoint on each pass of a binary search.

You may not need to use all the rows in the table.

(4)

high index	low index	midpoint

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1. Question 5 · part b · 4 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part b only.

(c) **Figure 7** shows the **Sales.txt** file. It stores sales information.

```
264,140,120,284,192
420,377,435,376,392
619,589,606,586,600
799,811,788,814,788
982,1007,1013,989,1009
```

Figure 7

A program is required to calculate and display:

- a subtotal for each line of sales
- a grand total for all the sales in the file.

Figure 8 shows the intended output from the program.

```
1000
2000
3000
4000
5000
Grand total: 15000
```

Figure 8

Open **Q05c** in the code editor.

Amend the code to produce the intended output.

You must use the structure and variables given in **Q05c** to complete the program.

Do not add any further functionality.

Save your code as **Q05cFINISHED** with the correct file extension for the programming language.

(8)

(Total for Question 5 = 14 marks)

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1. Question 5 · part b · 4 marks · Working and answer

Working and answer

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2. Question 4 · part b.ii · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b.ii only.

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4 Zak sells snacks at an after-school club.

(a) He wants a program that will hold details of the products he sells.

The program must:

- generate a five-character product code that:
  - starts with the first three letters of the product name
  - ends with a random number between 10 and 30
- display the product code followed by the product name.

Open **Q04a** in the code editor.

Write the program.

You must use the structure given in the file **Q04a** to complete the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(7)

(b) Zak plans to implement a binary search algorithm to search a table of products.

(i) Explain **one** advantage of a binary search compared to a linear search.

(2)

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2. Question 4 · part b.ii · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b.ii only.

(ii) **Figure 2** shows an algorithm for a binary search.

```
1    #    Initialise variables
2    SET productSearch TO " "
3    SET startPosition TO 1
4    SET midPosition TO 0
5    SET endPosition TO LENGTH(productList)
6    SET found TO False
7
8    #    Print prompts, take and check input from user
9    SEND "Enter the product code" TO DISPLAY
10   RECEIVE productSearch FROM (STRING) KEYBOARD
11   REPEAT
12       midPosition = (startPosition + endPosition) DIV 2
13       IF productList[midPosition] < productSearch THEN
14           startPosition = midPosition + 1
15       END IF
16       IF productList[midPosition] > productSearch THEN
17           endPosition = midPosition – 1
18       END IF
19   UNTIL productList[midPosition] = productSearch OR startPosition = endPosition
```

Figure 2

This binary search algorithm will be used to search the product list for the product code **str15**.

Complete the table to indicate the order in which the product codes will be examined by the algorithm.

Write the number 1 by the first product code to be examined, 2 by the second code to be examined, and so on.

(4)

Position in list	Product code	Order examined
1	ark11	
2	asp11	
3	bar13	
4	dri15	
5	mil19	
6	rib10	
7	str15	
8	tor16	

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P 6 6 4 9 0 A 0 1 2 1 6

2. Question 4 · part b.ii · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b.ii only.

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(iii) Zak has another list containing the names of five students who attend the after-school club.

Give the maximum number of names that would need to be examined by the binary search algorithm to determine whether a name appears in the list.

(1)

(iv) Give the name of an algorithm that could be used to sort a list.

(1)

(Total for Question 4 = 15 marks)

Working and answer

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3. Question 4 · part b.iii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part b.iii only.

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4 Zak sells snacks at an after-school club.

(a) He wants a program that will hold details of the products he sells.

The program must:

- generate a five-character product code that:
  - starts with the first three letters of the product name
  - ends with a random number between 10 and 30
- display the product code followed by the product name.

Open **Q04a** in the code editor.

Write the program.

You must use the structure given in the file **Q04a** to complete the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(7)

(b) Zak plans to implement a binary search algorithm to search a table of products.

(i) Explain **one** advantage of a binary search compared to a linear search.

(2)

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3. Question 4 · part b.iii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part b.iii only.

(ii) **Figure 2** shows an algorithm for a binary search.

```
1     #     Initialise variables
2     SET productSearch TO " "
3     SET startPosition TO 1
4     SET midPosition TO 0
5     SET endPosition TO LENGTH(productList)
6     SET found TO False
7
8     #     Print prompts, take and check input from user
9     SEND "Enter the product code" TO DISPLAY
10    RECEIVE productSearch FROM (STRING) KEYBOARD
11    REPEAT
12          midPosition = (startPosition + endPosition) DIV 2
13          IF productList[midPosition] < productSearch THEN
14                startPosition = midPosition + 1
15          END IF
16          IF productList[midPosition] > productSearch THEN
17                endPosition = midPosition – 1
18          END IF
19    UNTIL productList[midPosition] = productSearch OR startPosition = endPosition
```

Figure 2

This binary search algorithm will be used to search the product list for the product code **str15**.

Complete the table to indicate the order in which the product codes will be examined by the algorithm.

Write the number 1 by the first product code to be examined, 2 by the second code to be examined, and so on.

(4)

Position in list	Product code	Order examined
1	ark11	
2	asp11	
3	bar13	
4	dri15	
5	mil19	
6	rib10	
7	str15	
8	tor16	

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P 6 6 4 9 0 A 0 1 2 1 6

3. Question 4 · part b.iii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

(iii) Zak has another list containing the names of five students who attend the after-school club.

Give the maximum number of names that would need to be examined by the binary search algorithm to determine whether a name appears in the list.

(1)

.....

(iv) Give the name of an algorithm that could be used to sort a list.

(1)

.....

(Total for Question 4 = 15 marks)

Working and answer

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4. Question 2 · part b · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

2 Estelle is learning about sorting and searching algorithms.

(a) Estelle's friend has given her a partially completed program to carry out a binary search.

This pseudocode contains the logic required to complete the program.

```
11 RECEIVE item FROM (INTEGER) KEYBOARD
12 WHILE start <= end AND NOT found DO
13      SET middle TO (start + end) DIV 2
14      IF numberList[middle] = item THEN
15       SET found TO TRUE
16      ELSE
17       IF item < numberList[middle] THEN
18        end = middle - 1
19       ELSE
20        start = middle + 1
21       END IF
22      END IF
23      SET count TO count + 1
24 END WHILE
```

Open file **Q02a** in the code editor.

Amend the code to complete the program.

You **must** use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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4. Question 2 · part b · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

(b) **Figure 3** shows an array of names.

0	1	2	3	4	5	6	7
Yekatrina	Juan	George	Elija	Christine	Sangeetha	Melanie	Shu-fen

Figure 3

Explain why a binary search algorithm would not find 'Juan' in the array.

(4)

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6

P 6 9 2 9 8 A 0 6 1 6

4. Question 2 · part b · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

(c) A merge sort algorithm can be used to sort a list into ascending order.

The list 9, 1, 7, 6, 3, 5, 2, 8 needs to be sorted.

Complete the merge sort using the space provided.

(5)

9, 1, 7, 6, 3, 5, 2, 8

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1, 2, 3, 5, 6, 7, 8, 9

(Total for Question 2 = 19 marks)

4. Question 2 · part b · 4 marks · Working and answer

Working and answer

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